Tab 1

We're exploring how software systems can support facilitators, particularly less experienced facilitators, in understanding learner needs. Rather than using AI to teach students directly, we're currently investigating how learner interaction data can be transformed into actionable insights and recommendations for facilitators. The current direction is moving toward an AI-assisted facilitator decision support system that helps facilitators identify students who may need additional support, pacing adjustments, or different instructional approaches.

Current Problem

- Teacher shortages
- Inconsistent educational delivery
- Mixed ability classrooms
- Facilitators may not have teaching expertise

Current Idea

Students complete activities.

System collects:

- completion times
- retries
- hint usage
- progression trends
- engagement indicators

System then generates:

- learner summaries
- support indicators
- facilitator recommendations
    - Prompts for them to use to help students better

"How are you modelling learning and teaching?"

Question 1

How are you representing a learner?

Ask:

> In your agentic system, how do you represent a student? What information is used to model their learning process?

This could be hugely useful.

You're currently struggling with:

> What learner data actually matters?

Anthony may have already solved part of this.

Question 2

What signals indicate learning?

Ask:

> What indicators does your system use to determine whether learning is occurring?

This is almost exactly your problem.

Examples:

- task completion
- mistakes
- retries
- improvement over time
- confidence

Question 3

What level of abstraction works?

You're currently trying to figure out:

Student Activity
↓
???
↓
Facilitator Recommendation

Ask:

> How detailed is your learner model? Do you model individual behaviours, skills, competencies, or something else?

Question 4

How do you avoid educational assumptions?

This is probably the most important question.

Ask:

> How do you ensure the system doesn't make unsupported educational conclusions?

Because this is exactly what you've been worried about.

Question 5

What makes explanations useful?

Ask:

> When your system produces outputs, how do you make them interpretable to users?

You're likely going to need explainability for facilitators.

Question 6

What did you remove from scope?

This is a fantastic Masters-student question.

Ask:

> What ideas sounded interesting initially but ended up being too large or difficult to justify?

This might save you months.

What I'd love to learn if I were you

Honestly?

I'd spend most of the meeting trying to understand:

How does he model teaching?

Because that's where your project feels weakest right now.

You have lots of ideas about:

- dashboards
- AI
- analytics
- recommendations

But you're still missing a strong conceptual model of:

Student
↓
Learning Behaviour
↓
Interpretation
↓
Facilitator Action

If Anthony's system explicitly models the teaching-learning process, he may already have frameworks, papers, or design approaches that could help you enormously.

Benchmarking, OpenClaw

- Testing whether the system is reliable or not
- Testing Educational system with a student model
    - How to model accurately?
        - RAG → Retrieval Augmented Generation (LLM Database)
        - Use LLM to query data
        - Knowledge base → used to generate the database (brain of the student)
        - How did he get the data?
        - VCE Curriculum → Has report (with no. of students for each grade etc)
            - Secondary school (last year before uni)
        - Victoria → Australia Curriculum
    - Prompt Engineering
        - Too complex for alone
    - VCE Test → Exam (Australia and NZ standard)
    - Answering all the questions with the AI model
- Marking Agent
- Why Modelling Student Model
    - Gap can't really test how students will perform reliably
    - Used to test whether tests are working well / assess students effectively
    - Parameters → Correctness (multiple choice), okay with short answer questions
    - Using OpenAI → only prompts and responses
- Benchmarking
    - Agentic AI system → set goals and make step by step plans
    - Research with educational papers
        - OLM Open Learner Models → examine students learning
    - AI chat with students → doesn't
    - Educational System → interaction between human and system
        - Interacts with the student model and system

Assessing Students with Certain Tests

- Feedback from students
- Assess the students intermediate responses
- Assessing during learning → exam results are not best measure
- Each agents will talk to student → Need intermediate working (mathematics focus)

Facilitators

- Think about
- Research into
- Students → Assessment (How will I assess students?)
    - Feedback from intermediate learning
    - Combination of concepts → provide feedback to student
    -
- Facilitator Help → different materials (How will I help facilitators?)

Thursday morning → Wednesday afternoon NO

Focus on one subject and target group → LITERACY (fundamental) + older ish kids (more plausible / more assumptions can be made)

For students:

- Learner Readiness for Independent Learning
    - Reading Comprehension
    - Instruction Following
    - Persistence / Self - Determination
    - Help - Seeking / Feedback Reactions
    - Confidence

For facilitators:

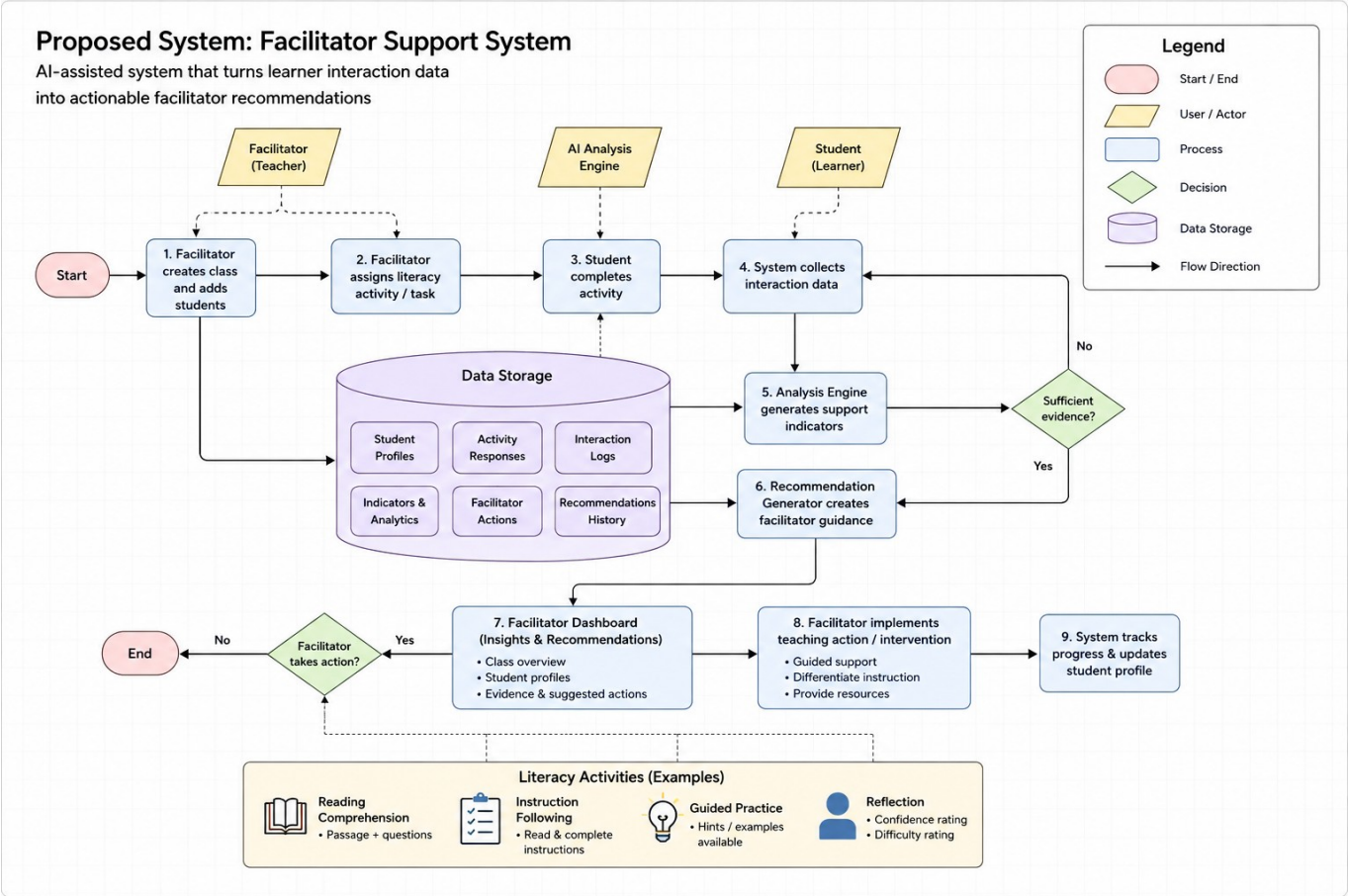
- Helping them understand whether a student doesn’t understand the content or understanding the instructions (reading comprehension and instruction following)

Success Criteria

Proposed Success Criteria:

- Assessing **four main areas** of the system:
  - Data Collection : Can meaningful learner behaviours be captured?
  - Learner Modelling : Can the system identify support needs?
  - Recommendation Quality : Can the system generate explainable facilitator guidance?
  - Decision Support : Can the system support facilitator decision making?
    - (include scope)

Criteria	Purpose	Measurement Method	Success Condition
Learner Data Collection	Ensure meaningful learner interactions are captured	Verify completion time, retries, hints, confidence, and accuracy are recorded correctly	100% of required interaction metrics stored
Behavioural Analysis	Ensure learner interaction data is correctly transformed into behavioural indicators	Run predefined learner scenarios	Expected indicators generated for all scenarios
Recommendation Generation	Ensure the system can produce facilitator guidance	Compare generated recommendations against expected outputs	Recommendations generated for all test scenarios
Recommendation Explainability	Ensure recommendations are supported by evidence	Verify every recommendation references observed behaviours	All recommendations contain supporting evidence
Facilitator Decision Support Coverage	Ensure the system addresses intended facilitator decisions	Map recommendations to facilitator needs	System supports all identified decision categories
Learner Progress Tracking	Ensure changes in learner behaviour can be monitored over time	Simulate multiple activity sessions	Profile updates reflect learner progression
Class-Level Analysis	Ensure facilitators can identify broader classroom trends	Create multi-student test dataset	System identifies class-wide patterns correctly
Transparency	Ensure the system avoids unsupported conclusions	Review generated outputs	No diagnoses or unsupported educational claims produced
End-to-End Functionality	Ensure the complete workflow functions correctly	Execute full workflow from activity completion to recommendation generation	Successful completion of all workflow stages



Purpose of MVP:

- To demonstrate that learner interaction data can be transformed into meaningful, explainable recommendations that support facilitator decision-making.
    - (Evaluates workflow, system design, concept rather than educational outcomes)

Research Question:

- Can a software system collect learner interaction data and present it in a way that helps facilitators understand student support needs?

Can learner interaction data be used to generate evidence-based recommendations that support facilitator decision-making?

Can a software system convert learner interaction data into evidence-linked recommendations for facilitator decision-making?

Camera / Eye Movement Ideas

For this project, I would frame camera tracking as a future optional support signal, not core MVP. Especially because this is Year 3-6, you would need strong consent, privacy, and probably ethics approval if using real students.

Useful signals you could track:

1) Question vs passage gaze
   Detect whether the student spends more time looking at the question, passage, or answer options.
   Example support indicator: “Student repeatedly returned to the question before answering.”
2) Looking away / attention breaks
   Track when the student looks away from the task for long periods.
   Useful as a low-confidence engagement signal, not as proof of understanding.
3) Question reread proxy
   If gaze repeatedly returns to the question area, that may indicate uncertainty about what to do.
4) Answer option scanning
   Track whether the student scans all answer options or fixates quickly on one.
   Could support recommendations like “Prompt student to compare all options.”
5) Calibration-light MVP
   For now, you could avoid actual eye-tracking and use webcam only as a concept: divide the screen into regions and estimate gaze direction roughly. Store only derived metrics, never raw video.

Best technical direction later:

Use browser webcam access with local-only processing.
Use approximate gaze/face landmarks, not identity recognition.
Store metrics like questionDwellMs, passageDwellMs, answerDwellMs, lookAwayCount.
Do not store video frames.

For the report, I’d say the MVP currently uses interaction-based proxies, and camera-based gaze tracking is a future extension for detecting whether difficulty is with instructions, attention, or content.

ML Approach

Machine Learning Predictions

- Question for facilitators: “How should I best support this student?”

Dimension	Meaning	Measurements	Example derived metric
Independence	Ability to complete tasks without external support	Hint count, passage revisits, time before first interaction, attempts	Independence Score (0–100)
Reading Comprehension	Understanding of the text	Question accuracy, question difficulty, time spent reading, incorrect answer patterns	Comprehension Score
Confidence Calibration	Relationship between perceived confidence and actual performance	Confidence ratings, accuracy	Calibration Index
Help-Seeking Behaviour	How support is used during learning	Hint timing, hint frequency, passage revisits after hints	Help-Seeking Profile
Task Processing	Ability to understand and follow instructions	Reading time before starting, first interaction delay, instruction/passage revisits	Task Processing Score

- Extensible
- Individual VS Class
- Decision Trees (Explainability)
- X - G boost ML model (gives slightly better results)

Stage 1 — Working pipeline

1. Finalise the five dimensions.
2. Define measurable features.
3. Implement event logging.
4. Build the feature extractor.
5. Implement rule-based classifications.
6. Display evidence and actions on the dashboard.

Stage 2 — ML prototype

1. Create labelled scenarios.
2. Export the feature dataset.
3. Train decision-tree models.
4. Evaluate against the rule baseline.
5. Save the models.
6. Create a prediction API.

Stage 3 — Explainable assistance

1. Return prediction confidence.
2. Extract the most influential evidence.
3. Generate cautious facilitator actions.
4. Show evidence, inference and action separately.

Stage 4 — Future validation

1. Gather ethically approved real data.
2. Replace synthetic labels with expert-reviewed labels.
3. retrain and recalibrate the models.

4. Study whether facilitators find outputs useful.

The final pipeline

1. Student completes literacy activity
2. Application records interaction events
3. Feature extractor calculates behavioural metrics
4. Rules or ML infer five support dimensions
5. Explanation layer selects supporting evidence
6. Recommendation layer maps support dimensions to possible actions
7. Dashboard presents:
    - observation
    - inference
    - evidence
    - suggested action
8. Facilitator makes the final decision